Simulation Exercise 1 - Deflection of Bimetal Switch

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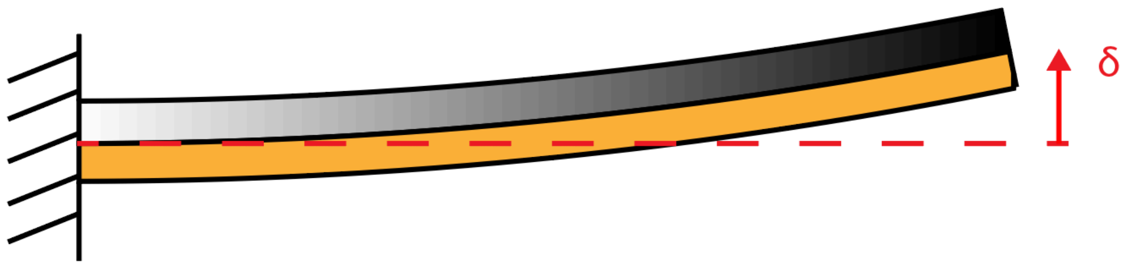

Simulation Exercise 1
FEM-modelling SE2860, 2025

Abstract

This study investigates the thermal deflection of a bimetallic switch composed of copper and steel using both analytical and numerical (FEM) methods in COMSOL Multiphysics. The analytical model predicted a deflection of 4.00 mm, while the numerical simulation yielded 4.08 mm, showing good agreement but indicating that the analytical model slightly underestimates deflection due to simplifying assumptions. The effects of different boundary conditions were also examined, revealing that fixed supports generate higher stresses and larger deflections than roller constraints. Finally, a thermostat simulation demonstrated that the switch closes at a heat influx of approximately 70000 W/m².

1. Problem background and definition

Bimetallic strips are composed of two bonded metals that expand at different rates when heated and thus bend in response to temperature changes (Axom, 2023). These strips, typically soldered or welded together to prevent separation, are widely used in temperature-sensitive devices like thermostats, refrigerators, and toasters, where precise deflection at specific temperatures is essential. Effective material selection and dimensional design are crucial to achieving the desired performance, and numerical tools such as COMSOL Multiphysics enable efficient simulation and optimization of these parameters. The behavior of bimetallic strips involves coupled heat transfer and thermal-induced solid mechanics: heat applied to the strip, often from below, drives conduction within the materials and convection at the surface. The governing physical phenomena are discussed in Sections 2.2 and 2.3.

2. Model definition

2.1 Geometry and materials

The top material of the switch is iron, while the material on the bottom is copper, the width (B) is constant for both materials. The two materials are joined using "Form Union" in COMSOL and the two strips were created using the block geometry. Half of the beam is modelled due to symmetry in the yz-plane. This is illustrated in Fig. 1.

Figure 1: *Dimensions and geometry of the switch*

Table 1: All parameters provided by the assignment description

Description	Parameter	Value	Unit
Length of bimetal	L	150	mm
Height of iron	h_1	1	mm
Height of copper	h_2	1	mm
Width	B	10	mm
Heat influx in a) and b)	$q_{in,ab}$	1000	W/m^2
Heat transfer coefficient	h	10	$W/(m^2K)$
Surrounding temperature	T_{Am}	293.15	K
Coefficient of thermal expansion for iron	α_1	12.2e-6	1/K
Coefficient of thermal expansion for copper	α_2	17e-6	1/K

Heat flux is denoted by q per thermodynamic convention (Ekroth & Granryd, 2021). Data in Table 1, is taken from the assignment description.

Table 2: All additional material parameters for steel and copper

Description	Parameter	Value	Unit
Young's modulus for iron	E_1	206	GPa
Young's modulus for copper	E_2	118	GPa
Specific heat capacity under constant pressure for iron	$C_{p,1}$	460	J/(kg·K)
Specific heat capacity under constant pressure for copper	$C_{p,2}$	390	J/(kg·K)
Density for iron	ρ_1	7850	kg/m^3
Density for copper	ρ_2	8940	kg/m^3
Poisons number for iron	υ_1	0.3	-
Poisons number for copper	υ_2	0.33	-
Thermal conductivity for Iron	k_1	55	$W/\ (m * K)$
Thermal Conductivity for Copper	k_2	390	$W/(m * K)$

Source: Alfredsson, 2014

In Table 2, 141312-00 steel and 5011-02 copper were selected for their thermal expansion coefficients, which closely match the values given in the assignment

2.2 Heat conduction in the bimetal

The two metals in the bimetal switch are assumed homogeneous and isotropic. The partial differential equation governing transient heat conduction is

$$\nabla \bullet (k\nabla T) + Q = \rho C_p \frac{d}{dt} T \quad . \qquad (1)$$

The definition of each variable is featured in the appendix. Initially, temperature everywhere is equal to the ambient temperature:

$$\text{Initial conditions: } (t = 0): \quad T = T_{Am} \qquad (2)$$

At the boundaries of the bimetal, the following apply:

$$\text{Boundary conditions:} \quad T = \overline{T} \ \text{or} \ -k\nabla T\cdot n = q_n \bullet n \qquad (3)$$

In the present application, heat is prescribed at certain boundaries and leaves the free surface by convection at other boundaries. For the latter, q_n then is expressed as

$$q_n = h\left(T - T_{Am}\right), \qquad (4)$$

Here, h is the heat transfer coefficient and T_{Am} is the ambient (room) temperature. Eq. 1 is solved in COMSOL using the finite element method (FEM), with heat flux applied to the bottom of the copper strip (Fig. 2, Table 1) and convective flux on the top of the iron strip to dissipate heat, as described by Eq. 3.

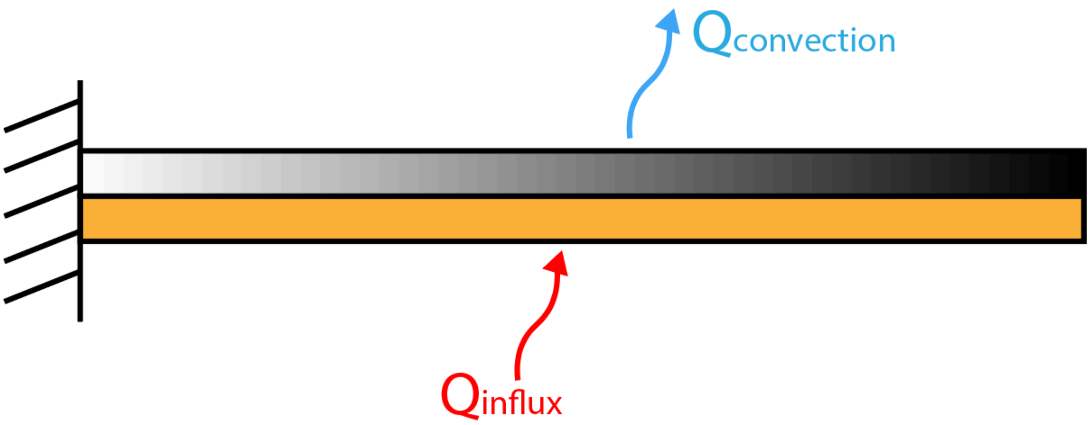

Figure 2: *Heat flux definition*

2.3 Solid mechanics equations

The two materials in the bimetal are assumed to be homogeneous, isotropic, linear elastic materials. The strains are defined by

$$\varepsilon = \frac{1}{2}\left(\nabla u + (\nabla u)^T\right), \qquad (5)$$

The constitutive equation, the relation between the stresses and the strains (Hooke's law), are given by

$$\sigma = 2G\varepsilon + \lambda\, tr(\varepsilon)I - 3K\alpha\Delta TI = C:\varepsilon_{el} \ with \ \varepsilon_{el} = \varepsilon - \varepsilon_{th},\ \varepsilon_{th} = 3K\alpha\Delta TI, \qquad (6)$$

It is assumed that the solid will deform under quasi-static conditions. In the absence of body forces, equilibrium is satisfied by

$$\nabla\sigma = 0. \qquad (7)$$

Combining equations (Eq. 5) to (Eq. 7), a set of partial differential equations is obtained, governing the solution to the solid mechanics problem, as

$$\nabla C : \left(\frac{1}{2}\left(\nabla u + (\nabla u)^T\right) - 3K\alpha\Delta T I\right) = 0. \qquad (8)$$

Initially it is assumed the solid are stress free:

$$\text{Initial conditions: } (t = 0): \quad u = 0 \qquad (9)$$

At the boundaries of the bimetal, the following apply:

$$\text{Boundary conditions: } \quad u = \bar{u} \ \text{ or } \ \sigma \bullet n = \bar{t} \qquad (10)$$

where $\bar{u}$ and $\bar{t}$, respectively, are prescribed at the boundaries. Eq. 8 is numerically implemented in COMSOL and solved by FEM. In COMSOL fixed boundary value a condition is applied in accordance with Fig 1, while all other sides are free, the boundary conditions are discussed further in Section 4: Results.

2.4 Analytic model for the temperature increase and deflection of the bimetal

A simple analytical model for the deflection of a bimetal switch under thermal loading was featured in a lecture by Jonas Faleskog 2025.

In solid mechanics, the free end of a beam has no moment nor normal force, as presented in Eq. 11 and 12.

$$N = 0 = \int_A \sigma dA = \int_{A1} \sigma_1 dA + \int_{A2} \sigma_2 dA \qquad (11)$$

$$M = 0 = \int_A z\sigma dA \qquad (12)$$

The strain, given by Eq. 13 has both a normal part ε_0 and a bending part z/R

$$\varepsilon = \varepsilon_0 + z/R \qquad (13)$$

The constitutive law was given by Hooke's law with thermoelastics, Eq. 14.

$$\sigma_i = E_i(\varepsilon_i + \alpha_i \Delta T) \quad (14)$$

The difference between ambient and the switch's temperature (ΔT) is calculated using the energy equation for the stationary state with the same effect in and out of the bimetal expressed in Eq. 15.

$$q_{in} A_{in} = h\Delta T A_{out} \qquad (15)$$

Since $A_{in} = A_{out}$ the equation is simplified to Eq. 16:

$$\Delta T = q_{in,ab}/h. \qquad (16)$$

The integral is simplified by $dA = Bdz$ since the stress is constant with respect to x. The limits are shown by Fig. 3.

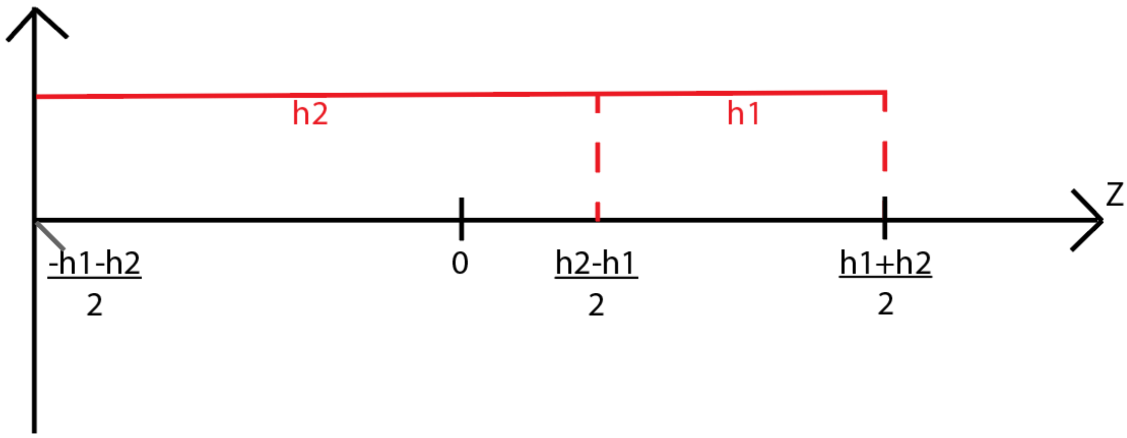

Figure 3: *Integration limits for normal force and moment integrals.*

Inserting Eq. 13 and Eq. 14 along with $dA = Bdz$ into Eq. 11 gives Eq. 17.

$$N = B\int_{-\frac{h_1+h_2}{2}}^{\frac{h_1+h_2}{2}} E_1\left(\varepsilon_0 + \frac{z}{R} - \alpha_1\Delta T\right)dz + B\int_{-\frac{h_1-h_2}{2}}^{\frac{h_1+h_2}{2}} E_2\left(\varepsilon_0 + \frac{z}{R} - \alpha_2\Delta T\right)dz = 0 \qquad (17)$$

The process is replicated for the moment, giving Eq. 18

$$M = B\int_{-\frac{h_1+h_2}{2}}^{\frac{h_2-h_1}{2}} E_1\left(\varepsilon_0 + \frac{z}{R} - \alpha_1\Delta T\right)z\,dz + B\int_{\frac{h_2-h_1}{2}}^{\frac{h_1+h_2}{2}} E_2\left(\varepsilon_0 + \frac{z}{R} - \alpha_2\Delta T\right)z\,dz = 0 \qquad (18)$$

The system of Eq. 17 and 18 are solved for R and ε_0. See Appendix C for attached matlab code. The deflection is calculated by integrating the curvature twice, giving Eq. 19

$$w = C_0 x - \frac{x^2}{2R} + C_1 \qquad (19)$$

Using the fixed constraint boundary condition $w(0) = \frac{dw(0)}{dx} = 0$, one gets $C_0 = C_1 = 0$, with $x = L$ giving the deflection at the end of the switch, Eq. 20.

$$\delta = \frac{L^2}{2R} \qquad (20)$$

This deflection is calculated using MATLAB.

3. Numerical model

3.1: Finite Element Model employed and Mesh Type

The FEM model was created in COMSOL Multiphysics 6.3. Since stress distribution in the cross section was of interest, a 2D system was insufficient, thus a 3D model was constructed. The "Thermal Stress, Solid" multiphysics package was selected, which included Solid Mechanics, Heat Transfer in Solids, and Thermal Expansion physics, meeting the model's requirements. A stationary study was used for cases a, b, c, while a time-dependent study was also applied for case c. For mesh type a quadrilateral hexahedral mesh was used, suitable for the rectangular geometry and simple application without cracks or complex features.

3.2: Mesh convergence study

The study used a *parametric sweep* in COMSOL, varying the parameter NumElement in the mesh distribution (Fig. 4). The maximum element size was reduced to 0.003 to avoid overly large elements; all other mesh settings were kept at COMSOL's default values. Results are shown in Section 4.1 and the convergence plot in Fig. 7.

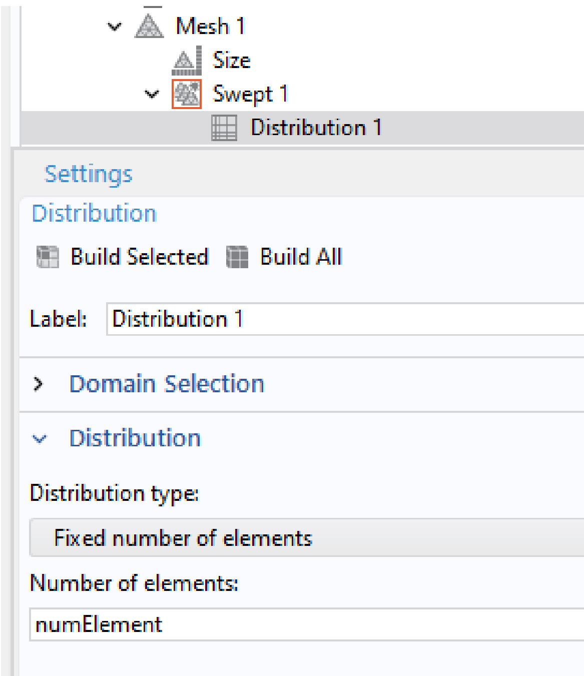

Figure 4: *Parameter used in the parametric sweep to investigate mesh convergence*

4. Results

The analyses were performed with a staggered approach, solving heat conduction then stress analysis. Three cases were considered: case a with fully fixed boundaries, case b with a fixed point and roller constraints, and case c, which adds a spring constraint to case b.

4.1 Fixed kinematic boundary conditions (a)

4.1.1: Boundary conditions

In case a, the left side was constrained using full fixed constraints, (Fig 5).

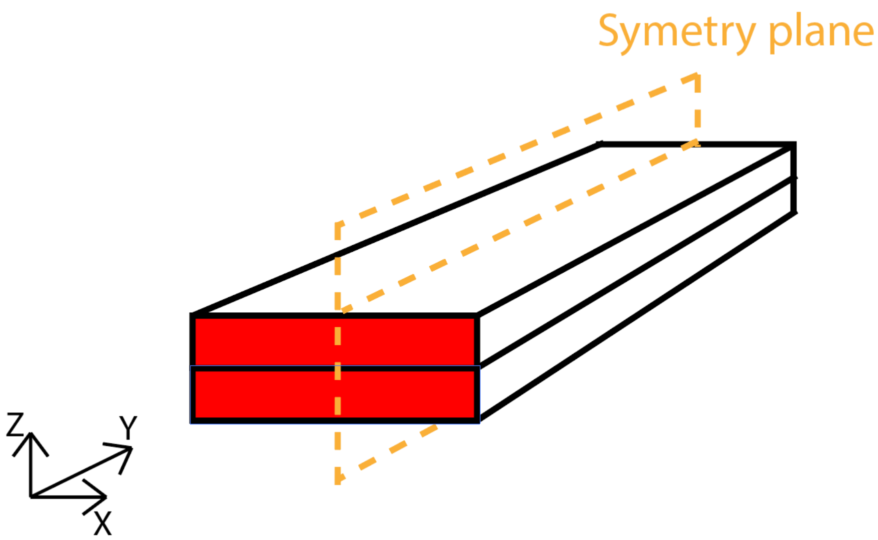

Figure 5: *Symmetry plane and fixed constraints (red)*

4.1.2: Results for case a

The deflection is shown through Fig. 6, where the displacement magnitude is shown. The maximum deflection at the free end is 4.08 mm.

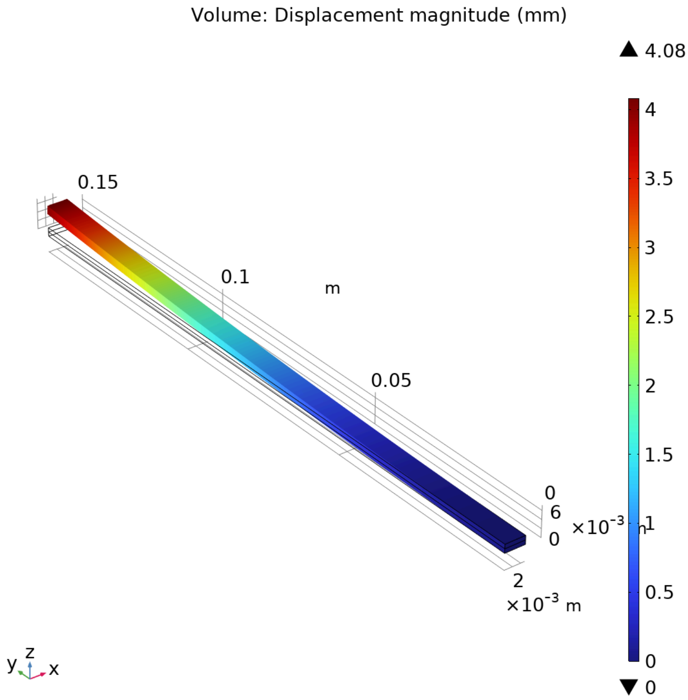

***Figure 6**: Displacement magnitude for case a*

Figure 7 shows the convergence plot from the parametric sweep with 5 to 20 layers of elements along the z-axis since this is the direction with highest temperature gradient, where the error is under 0.1% for all meshes. For the mesh convergence a point on the upper corner on the symmetry axis was selected for comparison, this point was used for all measurements or inputs where the end deflection was needed.

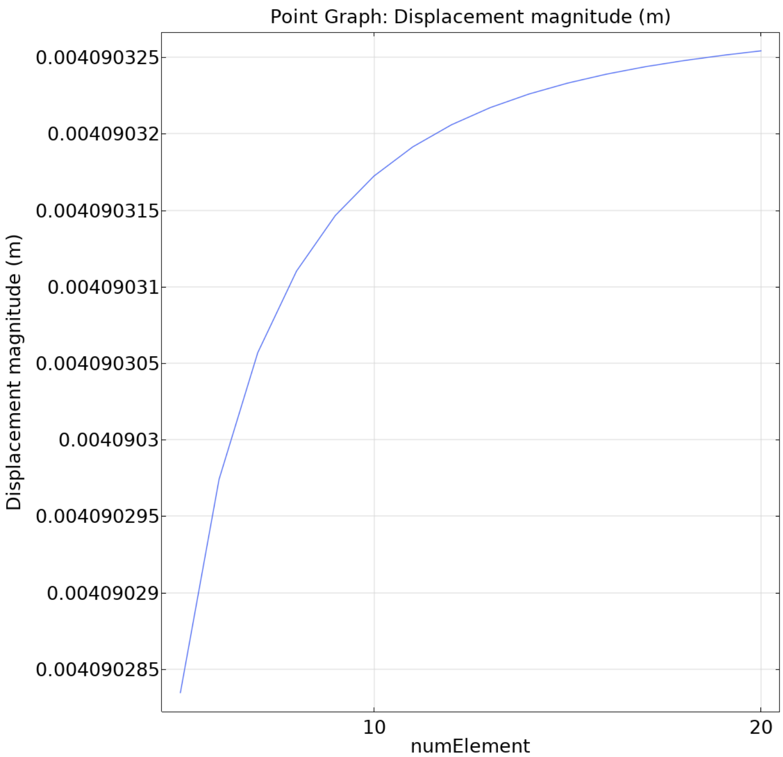

Figure 7: Displacement of the switch's end varies with increasing amount of elements in z-direction.

To maintain accuracy while minimizing computation time 10 elements were used for case a and b as seen in Fig. 8

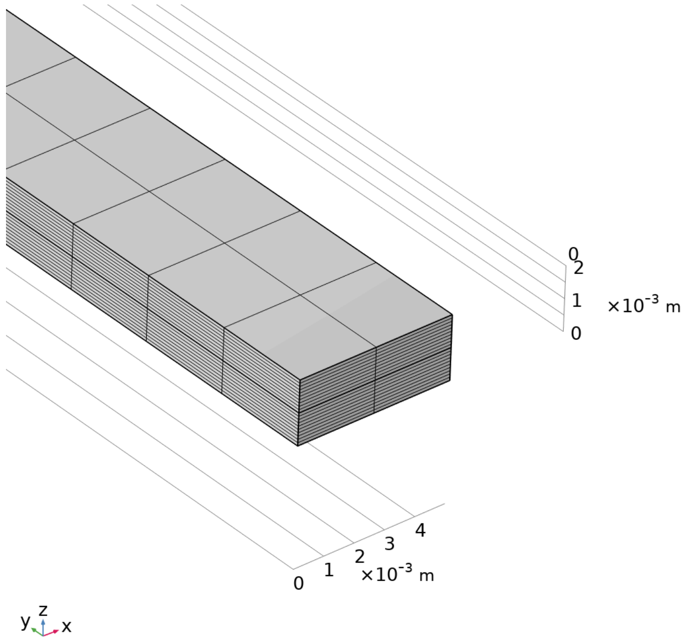

Figure 8: *Mesh used for case a and b. Maximum size 0.003 m and 10 layers.*

4.1.3: Comparison of numerical and analytical results

The analytical solution gives a deflection of 4.00 mm compared to the numerical that deflects 4.08 mm. This means that the analytical solution underestimates the deflection. The difference can be attributed to simplifications and idealizations made in the analytical model, for instance, Euler-Bernoulli beam theory is used, which states that plane sections remain plane during deformation, these idealizations means that the analytical solution is less stiff. Other assumptions like small deformations may also contribute to the difference.

4.2 Relaxed kinematic boundary conditions (b)

4.2.1: Boundary conditions

To allow in-plane deformation, only one fixed point was applied at the boundary, and on the symmetry plane (Figure 9), while the remaining support surface used roller constraints. This setup lets the surface expand freely without shear forces, unlike in case a.

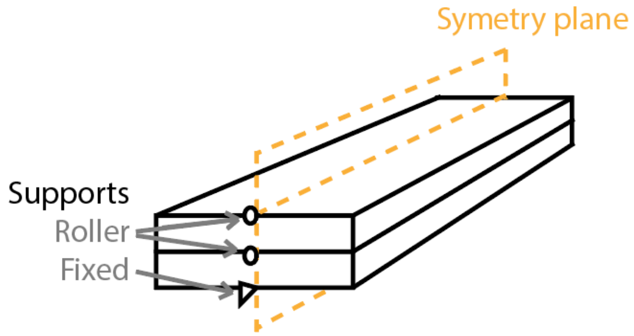

Figure 9: *Boundary condition at the supported surface.*

4.2.2: Results for case b

The displacement along the bimetallic switch for case b, the displacement is greatest at the free end. (Fig 10). From Fig. 10 the deflection of the edge is 3.9939 mm in positive z-direction.

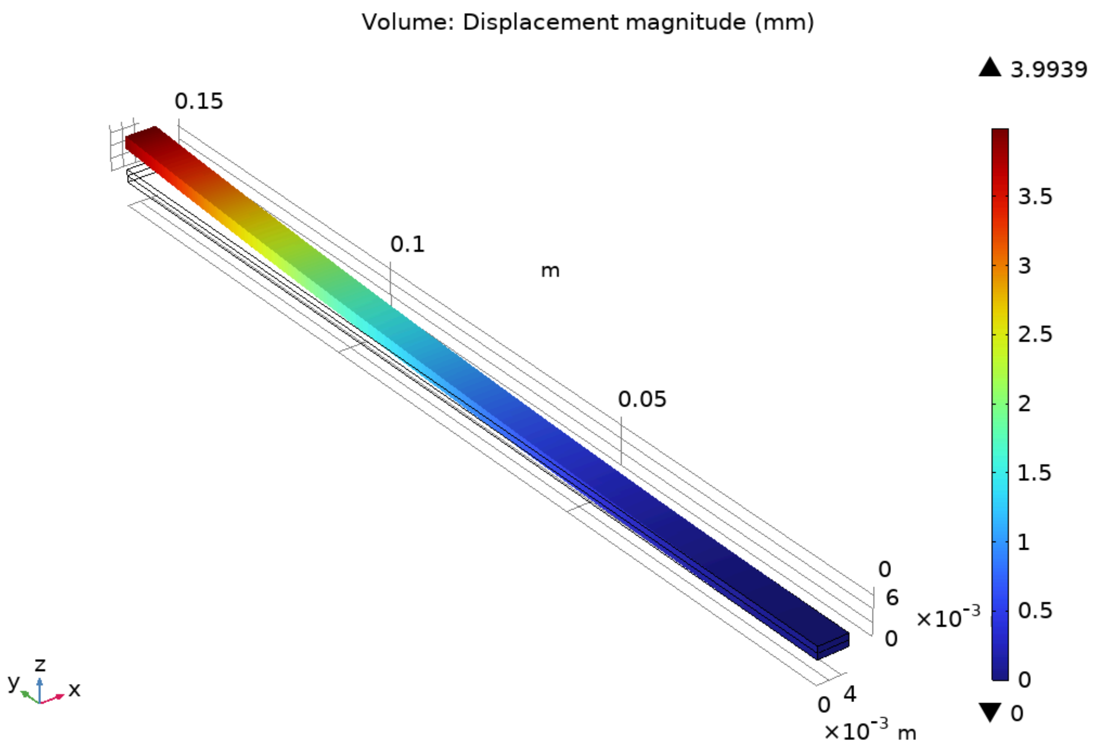

Figure 10: *Displacement of switch in case b.*

4.3 Simulation of a thermostat based on a bimetal switch (c)

In this case we have heating from the support and convection at the top and bottom of the switch, a stop is modelled at the end at a displacement $\delta_b/2$ as shown in Figure (11). The support had the same boundary conditions as in case (b).

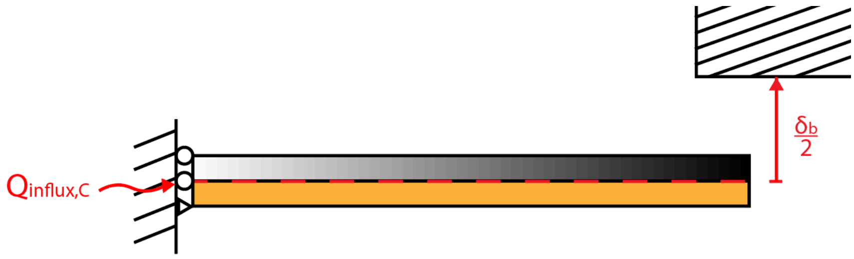

Figure 11: *Setup for case c.*

Since the gradient of the temperature now is in y-direction (Fig. 12) rather than the z-direction as in case (a) and (b), therefore the mesh used does not need as many elements in the z-direction as before, seen in Fig. 8. The mesh used instead is shown in Fig. 13 with more elements in the y-direction due to the temperature gradient.

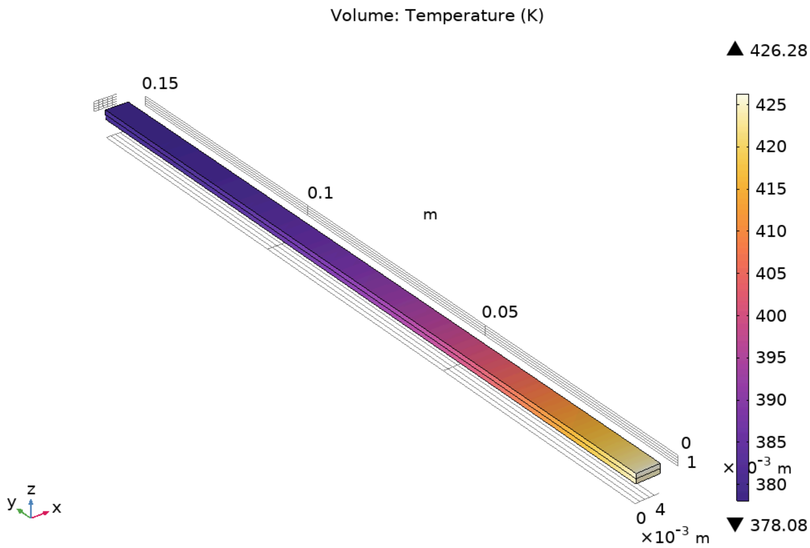

Figure 12: *Heat distribution of switch for case c during steady state*

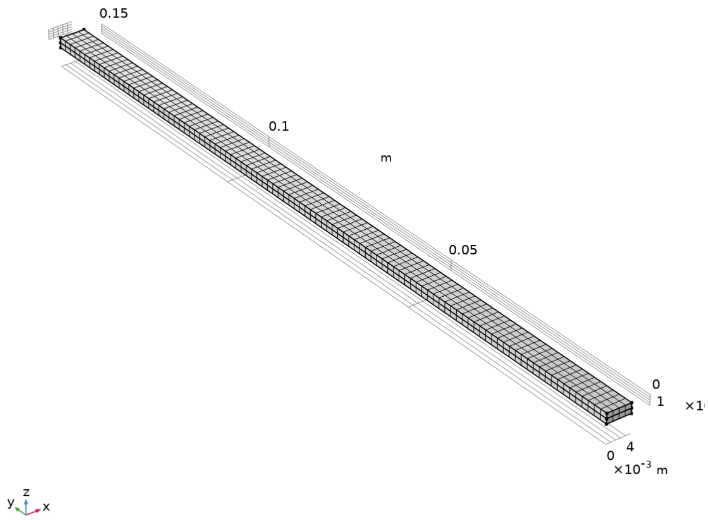

Figure 13: *Mesh of switch for case c. Maximum size 0.0015 m.*

A stationary study was performed with heat influx as a sweep parameter to determine the switch-closing point. The stop was omitted to allow the study, and the endpoint displacement was plotted against heat influx (Fig. 14).

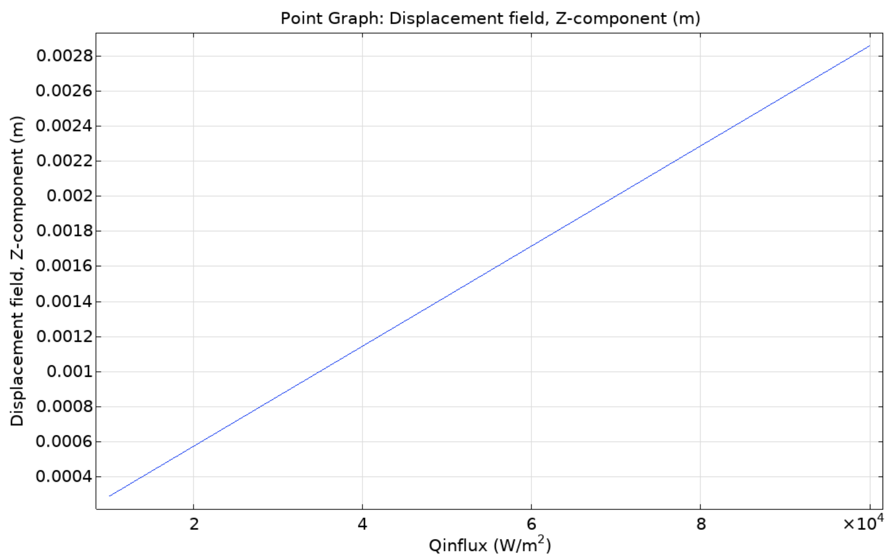

Figure 14: *Graph of displacement of switch in relation to heat influx.*

From this graph, the heating can be read at a displacement of $\delta_b/2 = 3.9969/2\ mm \approx 2\ mm$. This gives a heat influx of 70000 W/m^2.

To estimate the time when the switch closes, a time dependent study was used in COMSOL. The stop was modeled by the implementation of a bi-linerar spring in z-direction at a point on the end of the switch. To model a rigid stop, the stiffness of the spring was set as a stepfunction with the step initiated at $\delta_b/2$. For the simulation to converge without increasing the number of timesteps too much due to limited computational power, the stiffness stepfunction used had a smoothening 10^{-4}, a small starting stiffness set to 1 N/m and an end stiffness of 1000 N/m. For the spring to model the rigid stop realistically, the step should have gone from zero to infinity without any smoothing.

To get the time when the switch hits the contactpoint, a graph was plotted with the displacement in z-direction as a function of time as shown in Figure (15).

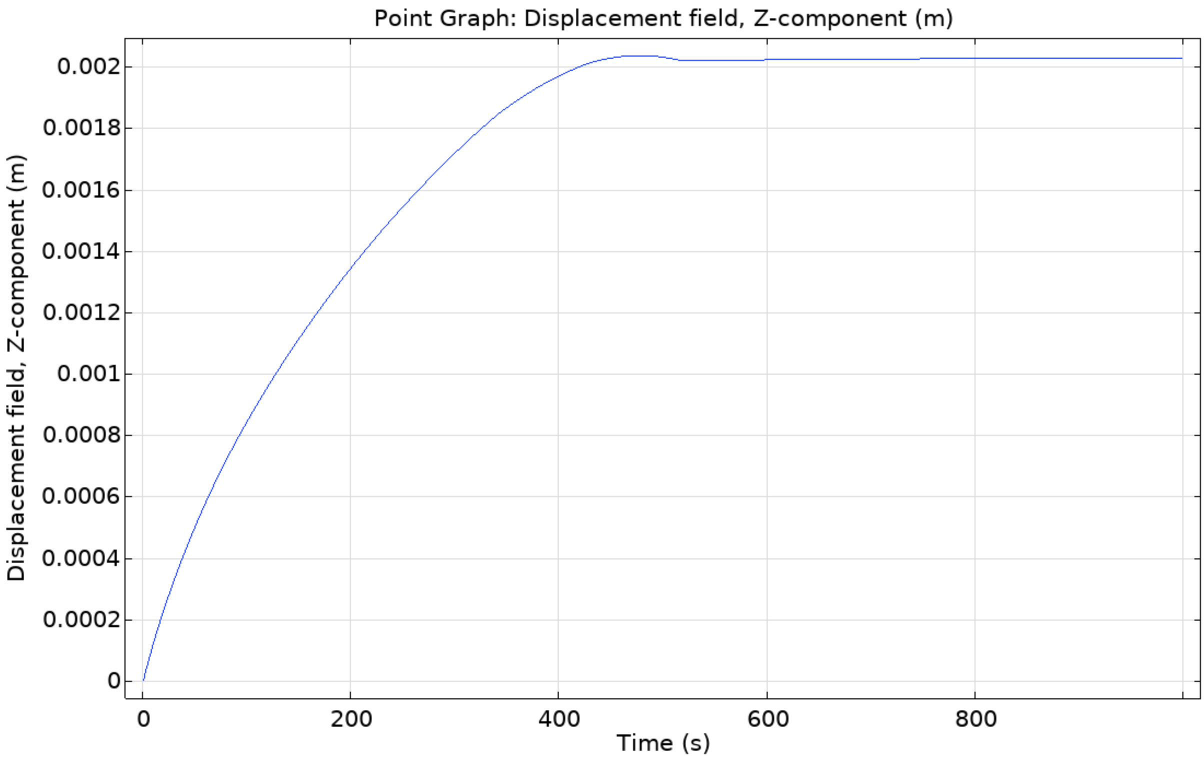

Figure 15: *Displacement of endpoint with heat influx 100000 W/m*^{2}

For the switch to close, heat influxes above 70000 W/m^2 were used and gave the results in Table (3). The total energy (E) flowed into the switch before closure is calculated by $E = q_{in,c} T_{close} A_{in}$, where T_{close} is the time the switch closes and A_{in} is the area with heat influx ($A_{in} = B(h_1 + h_2)$).

Table 3: Energy and time dependence of switch closing

Heat influx	Time of switch closing	Energy
100000 W/m^2	*443 s*	*886 J*
150000 W/m^2	*215 s*	*645 J*
200000 W/m^2	*143 s*	*572*

5. Discussion and conclusion

5.1: Comparison of the stress state in case a and case b

When comparing the stress states of cases a and b (Figs. 16 & 17), boundary conditions affect movement and stress distribution. In case a, the highest stress (around $6 * 10^8$ Pa) is localized at the fixed cross-section, while in case b, stress (around $8 * 10^7$ Pa) spreads along the copper and steel plane. Although case b's stress pattern is present in a, the high stress at the fixed constraints makes it difficult to distinguish between the plates.

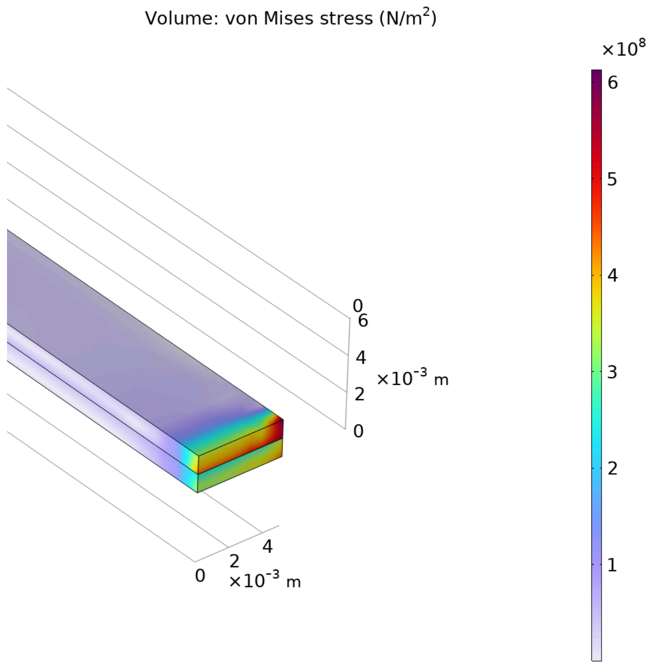

Figure 16: *Equivalent Von Mises stress for case a*

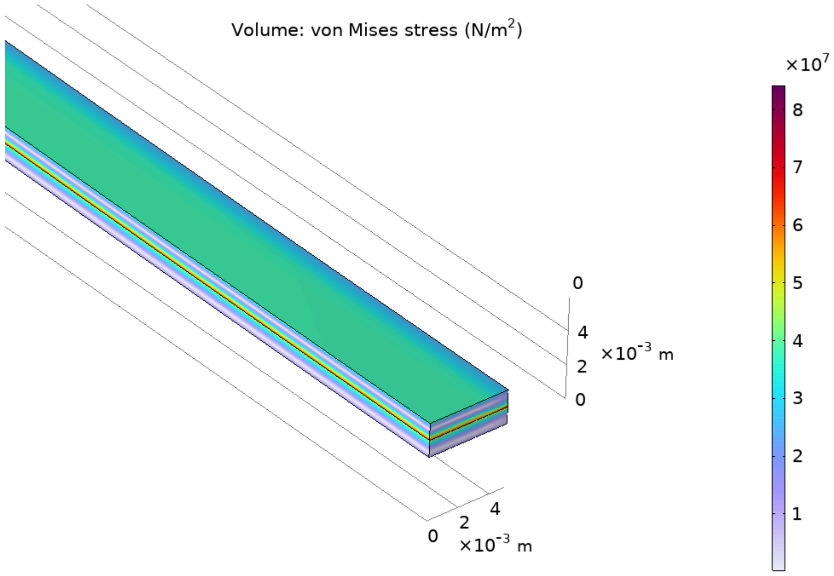

Figure 17: *Equivalent Von Mises stress for case b*

5.3: Conclusion:

- The analytical model predicted a deflection of 4.00 mm, while the numerical FEM model gave 4.08 mm, showing that the analytical model slightly underestimates deflection.
- When the boundary was fully fixed (case a), the structure experienced higher thermal stresses at the support and slightly larger deflection than case b that had 3.99 mm deflection.
- The numerical results converged after approximately 2 elements across the thickness, confirming mesh independence.
- For the thermostat simulation (case c), the switch closed at a heat influx of ~70,000 W/m², with closure time decreasing as heat flux increased.
- Overall, the study confirms that boundary conditions strongly influence both stress and deflection, and that analytical methods provide useful but slightly conservative estimates compared to detailed FEM simulations.

6. AI

6.1: ChatGPT use in the solution

ChatGPT was used to formulate the MATLAB code to calculate the analytical solution.

6.2: ChatGPT use in the report

In the report ChatGPT was used to proofread, correct spelling and formulation in the report, and also to shorten long paragraphs. It was used to generate readable equations, since Google Docs equation editor makes equations small. Lastly it was used to correctly cite sources in the reference section.

7. References

Axsom, T. (2023, March 30). The bimetallic strip explained. Fictiv. https://www.fictiv.com/articles/the-bimetallic-strip-explained

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Ekroth, I., & Granryd, E. (2021). *Tillämpad termodynamik* (2nd ed.). Studentlitteratur AB.

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8. Appendix

A. Plots of case (c)

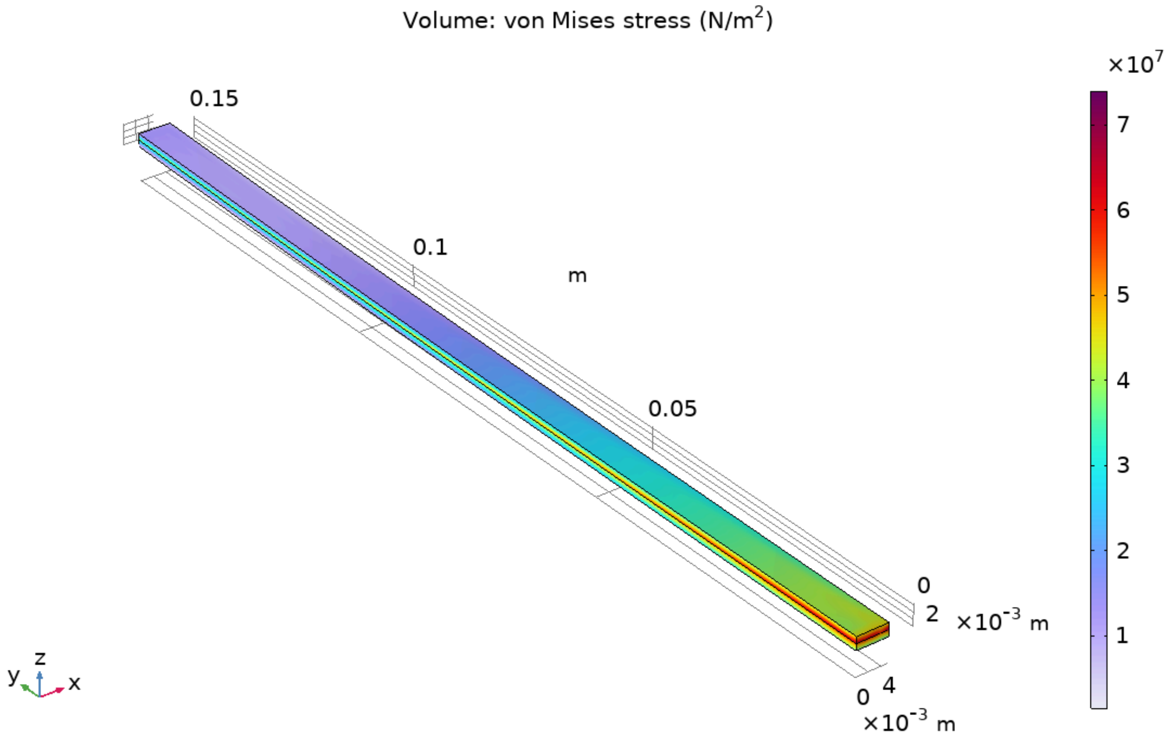

Figure A1: *Von Mises stress of beam for case c.*

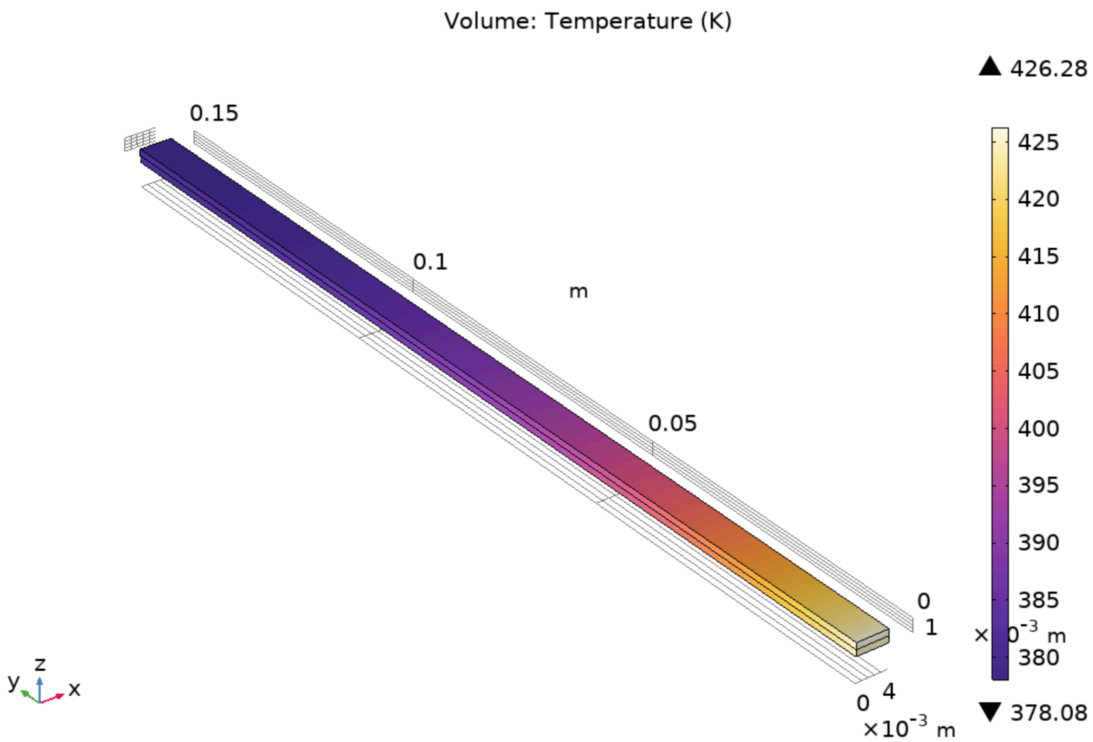

Figure A2*: Heat flux of switch for case c.*

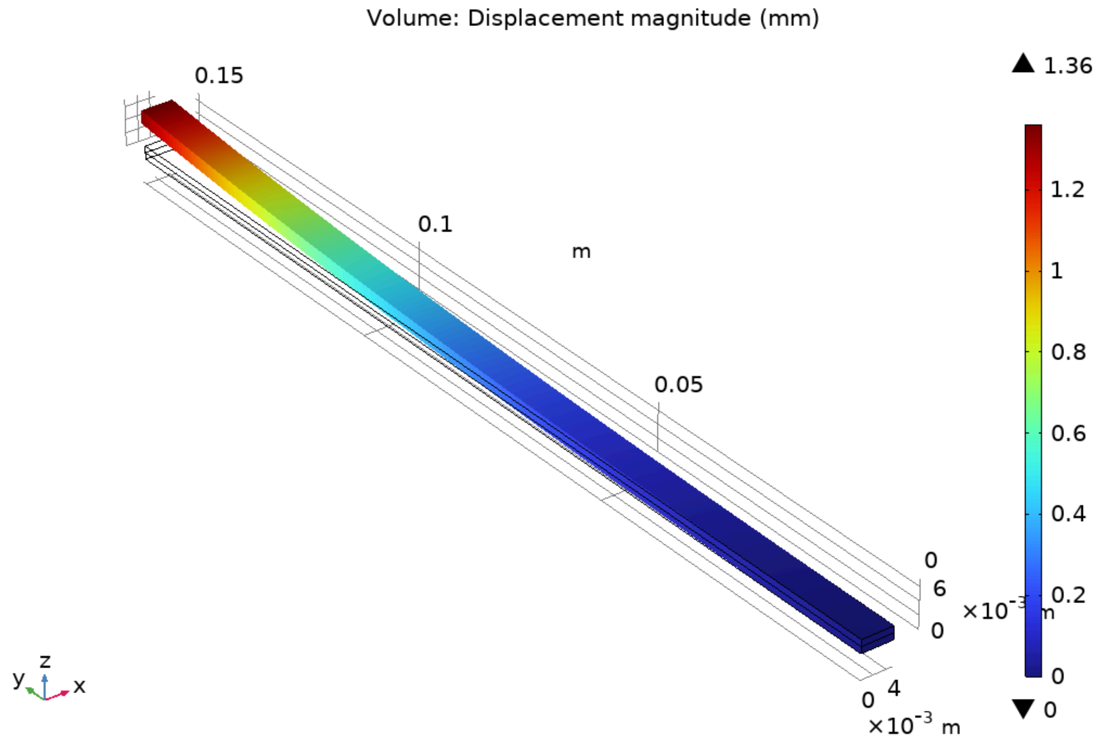

Figure A3: *Displacement magnitude of switch for case c.*

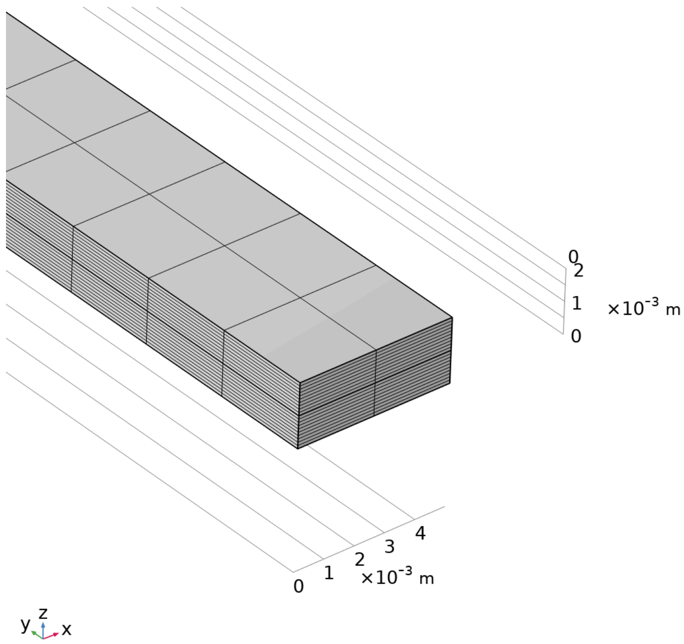

Figure A4*: mesh used in b with 10 layers*

B. Matlab code for analytical deflection in (a)

```
% Materialparametrar
E1 = 206e9;     % [Pa] Elasticitetsmodul lager 1 (t.ex. stål)
E2 = 118e9;     % [Pa] Elasticitetsmodul lager 2 (t.ex. koppar)
alpha1 = 12e-6;  % [1/K] Värmeutvidgningskoefficient lager 1
alpha2 = 17e-6;  % [1/K] Värmeutvidgningskoefficient lager 2
L = 0.15;
b=0.01;
Qin=1000;
h=10;
% Geometri
h1 = 1e-3;      % [m] tjocklek lager 1
h2 = 1e-3;      % [m] tjocklek lager 2
% Temperaturändring
dT = Qin/h;       % [K] temperaturökning
% ---- Beräkning av krökning ----
num = -6 * dT * E1 * E2 * h1 * h2 * (alpha1 - alpha2) * (h1 + h2);
den = (E1^2 * h1^4) + (4*E1*E2*h1^3*h2) + (6*E1*E2*h1^2*h2^2) ...
   + (4*E1*E2*h1*h2^3) + (E2^2*h2^4);
invR = num / den;     % [1/m] krökning = 1/R
R = 1 / invR;         % [m] radie av krökning
num_eps0 = dT * ( ...
   E1^2 * alpha1 * h1^4 + ...
   3*E1*E2 * alpha1 * h1^2 * h2^2 + ...
   4*E1*E2 * alpha1 * h1 * h2^3 + ...
   4*E1*E2 * alpha2 * h1^3 * h2 + ...
   3*E1*E2 * alpha2 * h1^2 * h2^2 + ...
   E2^2 * alpha2 * h2^4 );
den_eps0 = E1^2 * h1^4 + 4*E1*E2*h1^3*h2 + 6*E1*E2*h1^2*h2^2 ...
        + 4*E1*E2*h1*h2^3 + E2^2*h2^4;
eps0 = num_eps0 / den_eps0;  % [-] töjning
delta = L^2/(2*R);
fprintf('Krökning 1/R = %.3e [1/m]\n', invR);
fprintf('Radie R = %.3f [m]\n', R);
fprintf('Utböjning delta = %.3f [m]\n', delta);
% Gränser för de två lagren:
z1_lower = -(h1 + h2)/2;
z1_upper = (h2 - h1)/2;
z2_lower = z1_upper;
z2_upper = (h1 + h2)/2;
% --- Sigma-uttryck för respektive lager ---
sigma1 =@(z) E1 * (eps0 + z/R - alpha1*dT);
sigma2 =@(z) E2 * (eps0 + z/R - alpha2*dT);
sigma1z =@(z) E1 * (eps0 + z/R - alpha1*dT).*z;
sigma2z =@(z) E2 * (eps0 + z/R - alpha2*dT).*z;
% --- Integraler för normalkraft N ---
N1 = b * integral(sigma1, z1_lower, z1_upper);
N2 = b * integral(sigma2, z2_lower, z2_upper);
N = N1 + N2;
% --- Integraler för moment M ---
M1 = b * integral(sigma1z, z1_lower, z1_upper);
M2 = b * integral(sigma2z, z2_lower, z2_upper);
M = M1+M2;
```

C.

Table C1: Variable explanations for section 2.2-2.3

Variable	Explanation	Unit
T	Temperature	K
Q	Heat supply per unit volume	W/m^3
k	Thermal conductivity	W/(m·K)
ρ	Density	kg/m^3
C_p	Specific heat capacity	J/(kg·K)
∇	Gradient operator	1/m
q_n	Heat flux across the boundary in the direction of **n**.	W/m^2
$\boldsymbol{\varepsilon}$	Strain tensor	-
$\mathbf{u}$	Displacement vector.	m
$\boldsymbol{\sigma}$	Stress tensor	Pa = N/m^2
ε_{el}	Elastic strain tensor	-
ε_{th}	Thermal strain tensor	-
ΔT	Temperature change,	Kelvin
$\mathbf{I}$	Unity tensor	-
G	Shear modulus,	Pa = N/m^2
K	Bulk modulus	Pa = N/m^2

λ	Lamé's constant	Pa = N/m²
E	Elastic modulus	Pa = N/m²
ν	Poisson's ratio.	-
$\overline{T}$	Prescribed temperature	Kelvin
$\mathbf{n}$	Outward normal to the surface	-

D. Temperature:

Figure 8 shows the temperature distribution, ranging from 393.165 K to 393.150 K. The highest temperature occurs at the lower part where heat is applied and gradually decreases toward the iron section that conducts the heat away.

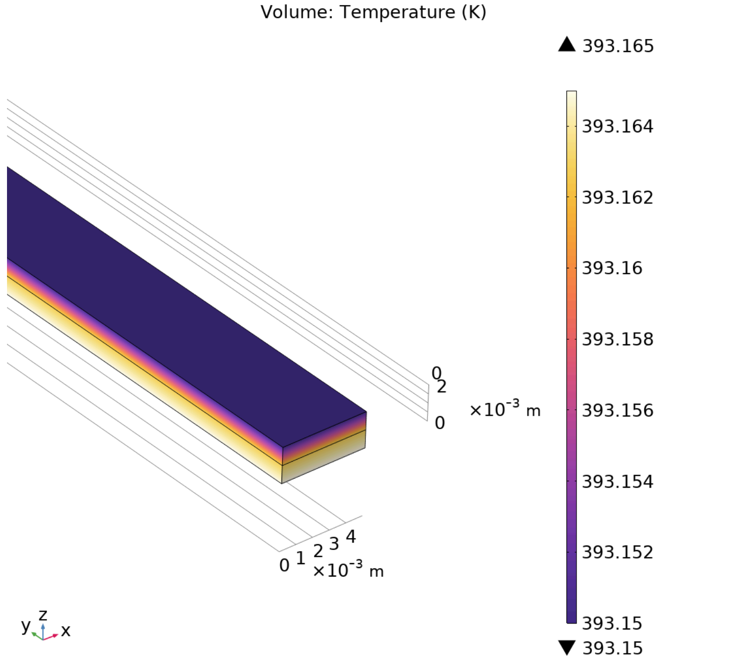

Figure D1: *Illustrates the distribution of heat flux in the beam for case a and b with small difference in temperature variance.*

Fig. D2 shows how the temperature varies for part b. It remains unchanged compared to case a.

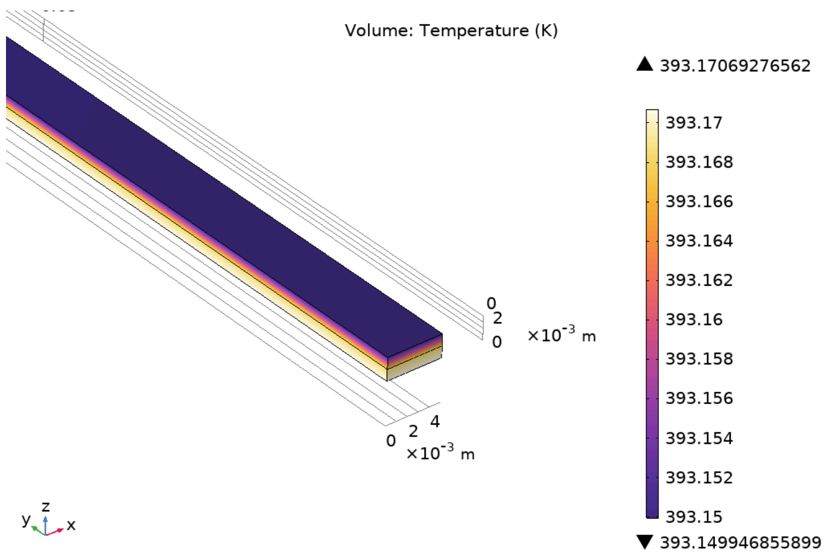

Figure D2: *Heat flux in beam, case b, displaying maximum and minimum temperature.*